

Module 5: Genetics, Human Inheritance and Cancer

Comprehensive Identification & Multiple-Choice Reviewer — 60 Items

Name: _____

Date: _____ Score: _____

INSTRUCTIONS: Read each statement carefully. Identify the correct term or concept described in the blank (____). Choose the letter of the correct answer from the choices provided (a, b, c, or d) and write it on the blank provided. An exhaustive Answer Key with detailed rationales is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE (60 ITEMS)

- _____ 1. What biological discipline is defined as the scientific study of heredity in general and of genes in particular?
a.) Cytology b.) Genomics c.) Genetics d.) Histology
- _____ 2. Gregor Mendel referred to the inherited units passed onto descendants as 'factors', which are now scientifically known as what?
a.) Enzymes b.) Genes c.) Histones d.) Chromatids
- _____ 3. What term describes the alternative or different forms of the same gene located at the same locus on homologous chromosomes?
a.) Histones b.) Codons c.) Alleles d.) Autosomes
- _____ 4. An individual that possesses a pair of identical alleles for a particular gene (such as AA or aa) is described as what?
a.) Homozygous b.) Hemizygous c.) Heterozygous d.) Polygenic
- _____ 5. An individual that possesses two different alleles for a particular gene (such as Aa) is described as what?
a.) Monohybrid b.) Homozygous c.) Heterozygous d.) Aneuploid
- _____ 6. In a heterozygous condition, what term describes the allele that masks the other allele and is phenotypically expressed?
a.) Incomplete b.) Codominant c.) Dominant d.) Recessive
- _____ 7. In a heterozygous condition, what term describes the allele whose phenotypic expression is masked or suppressed?
a.) Carrier b.) Dominant c.) Codominant d.) Recessive
- _____ 8. At the molecular level, what does a recessive allele typically produce compared to a dominant allele?
a.) A nonfunctional protein or no protein at all b.) A toxic protein c.) An identical duplicate protein d.) A structural lipid
- _____ 9. Which Mendelian law states that allele pairs separate during gamete formation so that each gamete carries only one allele for each gene?
a.) Law of Polygenic Inheritance b.) Law of Segregation c.) Law of Independent Assortment d.) Law of Dominance
- _____ 10. Which Mendelian law states that each pair of alleles located on one kind of chromosome separates into gametes independently of alleles on different chromosomes?
a.) Law of Continuous Variation b.) Law of Segregation c.) Law of Independent Assortment d.) Law of Dominance
- _____ 11. What genetic term refers to the precise genetic makeup or combination of alleles present within an individual?
a.) Phenotype b.) Karyotype c.) Genotype d.) Pedigree
- _____ 12. What term describes the observable physical, physiological, or biochemical traits expressed by an individual?

a.) Allele frequency b.) Genotype c.) Genome d.) Phenotype

_____ 13. What diagrammatic tool is used in genetics to calculate and predict the probabilities of offspring genotypes and phenotypes from parental gametes?

a.) Chomsky grid b.) Punnett square c.) Pedigree chart d.) Karyotype diagram

_____ 14. If a homozygous dominant individual (AA) crosses with a homozygous recessive individual (aa), what will be the genotype of all their offspring?

a.) 25% AA, 75% aa b.) 50% AA, 50% aa c.) 100% Heterozygous (Aa) d.) 100% Homozygous dominant (AA)

_____ 15. When two heterozygous individuals (Aa x Aa) mate, what is the expected genotypic ratio among their offspring?

a.) 100% Aa b.) 25% AA, 50% Aa, 25% aa c.) 50% AA, 50% aa d.) 75% Aa, 25% aa

_____ 16. When two heterozygous individuals (Aa x Aa) mate, what is the expected phenotypic ratio for a trait following complete dominance?

a.) 100% dominant b.) 75% dominant, 25% recessive c.) 50% dominant, 50% recessive d.) 25% dominant, 75% recessive

_____ 17. What term is given to a heterozygous individual who displays a normal/dominant phenotype but carries a recessive allele for a genetic disorder?

a.) Vector b.) Proband c.) Mutant d.) Carrier

_____ 18. What is a chart constructed to illustrate genetic relationships, ancestral lineages, and inheritance patterns among individuals in an extended family?

a.) Punnett square b.) Cladogram c.) Pedigree d.) Karyogram

_____ 19. What type of non-Mendelian inheritance occurs when both alleles are simultaneously and clearly apparent in the heterozygous phenotype?

a.) Incomplete dominance b.) Polygenic inheritance c.) Codominance d.) Complete dominance

_____ 20. What pattern of inheritance occurs when a heterozygous phenotype is intermediate between the two homozygous phenotypes (e.g., wavy hair from curly and straight)?

a.) Incomplete dominance b.) Sex-linkage c.) Pleiotropy d.) Codominance

_____ 21. What genetic scenario occurs when there are three or more distinct alleles for a single gene existing within a population, as exemplified by the ABO blood group system?

a.) Polygenic inheritance b.) Multiple alleles c.) Gene duplication d.) Epistasis

_____ 22. What form of inheritance occurs when two or more distinct genes govern the expression of a single continuous phenotypic trait such as height, skin color, or eye color?

a.) Incomplete dominance b.) Monogenic inheritance c.) Multiple alleles d.) Polygenic inheritance

_____ 23. Traits whose phenotypic expression depends on both the presence of specific alleles and the levels of circulating sex hormones (such as male-pattern baldness) are called what?

a.) Sex-influenced traits b.) Mitochondrial traits c.) Sex-linked traits d.) Autosomal recessive traits

_____ 24. In the human genetic traits inventory, which earlobe phenotype is dominant over attached earlobes?

a.) Lobeless b.) Pointed earlobes c.) Free earlobes d.) Attached earlobes

_____ 25. In human traits, is having a widow's peak hairline dominant or recessive to a straight hairline?

a.) Dominant b.) Polygenic c.) Codominant d.) Recessive

_____ 26. In human hand genetics, is the presence of six fingers (polydactyly) dominant or recessive to having five fingers?

a.) Recessive b.) Dominant c.) Sex-linked d.) Codominant

_____ 27. When clasping hands with thumbs crossed, which thumb placement is genetically dominant in humans?

a.) Parallel thumbs b.) Interlocked thumbs c.) Right thumb over left thumb d.) Left thumb over right thumb

_____ 28. In human facial genetics, is the presence of facial dimples dominant or recessive?

a.) Recessive b.) Sex-influenced c.) Incomplete dominant d.) Dominant

_____ 29. In tongue genetics, is the ability to roll the side edges of the tongue upward dominant or recessive?

a.) Incompletely penetrant b.) Dominant c.) Recessive d.) Codominant

_____ 30. In toe anatomy, which condition is genetically dominant regarding toe length in humans?

a.) Second toe longer than first toe b.) First toe longer than second toe c.) Equal length toes d.) Third toe longer than first toe

_____ 31. Which prenatal diagnostic procedure involves sampling and analyzing amniotic fluid and floating fetal cells from within the pregnant mother to detect genetic disorders?

a.) Ultrasound imaging b.) Chorionic villi sampling (CVS) c.) Amniocentesis d.) Fetoscopy

_____ 32. Which prenatal diagnostic technique samples cells directly from the placental projections to detect chromosomal and genetic abnormalities in a developing fetus?

a.) Amniocentesis b.) CT scanning c.) PET scanning d.) Chorionic villi sampling (CVS)

_____ 33. What fundamental biological characteristic is universally shared by all forms of cancer?

a.) Uncontrolled cell division b.) Bacterial infection c.) Programmed cell death d.) Decreased protein synthesis

_____ 34. What is an abnormal mass of tissue resulting from uncontrolled cell division, also known medically as a neoplasm?

a.) Tumor b.) Cyst c.) Histone d.) Polyp

_____ 35. What type of tumor is enclosed within a capsule of connective tissue and does not invade surrounding tissues or spread to distant sites?

a.) Carcinoma in situ b.) Benign tumor c.) Malignant tumor d.) Metastatic tumor

_____ 36. What type of tumor lacks a capsule, invades surrounding tissues, can spread throughout the body, and is properly classified as cancerous?

a.) Adenoma b.) Dysplastic cyst c.) Benign tumor d.) Malignant tumor

_____ 37. What term defines the migration and spread of cancer cells from their primary tumor site to distant locations throughout the body?

a.) Apoptosis b.) Metastasis c.) Angiogenesis d.) Dysplasia

_____ 38. What term describes the precancerous histological state marked by abnormal changes in cell shape, enlarged nuclei, and disorganized tissue architecture?

a.) Dysplasia b.) Carcinoma in situ c.) Hyperplasia d.) Metastasis

_____ 39. What term refers to a localized malignant tumor that has reached a critical mass of roughly 1 million cells but has not yet penetrated the basement membrane?

a.) Carcinoma in situ b.) Invasive carcinoma c.) Dysplasia d.) Benign neoplasm

_____ 40. What process is initiated when tumor cells secrete chemical signals that stimulate the formation and ingrowth of new blood vessels?

a.) Metastasis b.) Apoptosis c.) Cytokinesis d.) Angiogenesis

_____ 41. What is the physiological process of programmed cell death that eliminates cells with severe, irreparable genetic damage?

a.) Apoptosis b.) Necrosis c.) Mitosis d.) Dysplasia

_____ 42. What category of cellular genes acts as the cell's damage control system by detecting DNA damage and halting division or initiating repair?

a.) Tumor suppressor genes b.) Histone genes c.) Proto-oncogenes d.) Oncogenes

_____ 43. Normal regulatory genes that stimulate cell division in response to growth factors under controlled conditions are called what?

a.) Oncogenes b.) Tumor suppressor genes c.) Histones d.) Proto-oncogenes

_____ 44. What is a mutated, cancer-causing form of a proto-oncogene that drives continuous, uncontrolled cell division without normal external growth signals?

a.) Oncogene b.) Tumor suppressor gene c.) Allele d.) Autosome

_____ 45. Which two types of immune cells normally patrol the body and directly destroy emerging cancerous cells?

a.) Erythrocytes and platelets b.) Neutrophils and eosinophils c.) Natural killer cells and cytotoxic T cells d.) B cells and basophils

- _____ 46. Any environmental, chemical, physical, or biological agent that promotes genetic mutations and fosters the development of cancer is termed a what?
a.) Allergen b.) Histone c.) Carcinogen d.) Pathogen
- _____ 47. Which virus is a well-established biological cause of cervical, anal, and throat cancers in humans?
a.) Human Papilloma Virus (HPV) b.) Influenza virus c.) Rhinovirus d.) Norovirus
- _____ 48. Which two viral pathogens that cause chronic liver inflammation are established etiologic agents of hepatocellular carcinoma (liver cancer)?
a.) Hepatitis B and Hepatitis C viruses b.) Cytomegalovirus and Rabies c.) Rotavirus and Adenovirus d.) Hepatitis A and Hepatitis E viruses
- _____ 49. Which herpesvirus family member is causally linked to mononucleosis, Burkitt lymphoma, and nasopharyngeal carcinoma?
a.) Hepatitis A virus b.) Human Papilloma Virus c.) Varicella Zoster virus d.) Epstein-Barr virus (EBV)
- _____ 50. What type of environmental radiation, commonly emitted by the sun and tanning beds, damages cellular DNA and increases the risk of skin cancers?
a.) Microwave radiation b.) Infrared radiation c.) Radio waves d.) Ultraviolet (UV) radiation
- _____ 51. What diagnostic procedure involves extracting a small tissue specimen via needle or minor surgery for microscopic evaluation by a pathologist?
a.) Amniocentesis b.) Endoscopy c.) Angiography d.) Biopsy
- _____ 52. What diagnostic blood tests detect specific proteins or chemicals synthesized by tumor cells or produced by body tissues in response to a neoplasm?
a.) Complete blood count (CBC) b.) Hematocrit test c.) Tumor marker tests d.) Electrolyte panel
- _____ 53. Which bodily secretion is analyzed cytologically or genetically for malignant cells when screening for lung cancer?
a.) Feces b.) Sputum c.) Urine d.) Saliva
- _____ 54. Testing a patient's urine for atypical cells and specific mutational markers is primarily used to detect which type of malignancy?
a.) Colon cancer b.) Lung cancer c.) Breast cancer d.) Bladder cancer
- _____ 55. Analysis of fecal samples for occult blood and exfoliated mutant DNA markers is routinely employed to screen for which cancer?
a.) Bladder cancer b.) Brain cancer c.) Colon cancer d.) Lung cancer
- _____ 56. Which cancer treatment modality utilizes high-energy beams directed at malignant tissue to cause catastrophic double-strand DNA breaks and trigger apoptosis?
a.) Chemotherapy b.) Immunotherapy c.) Surgery d.) Radiation therapy
- _____ 57. Which systemic cancer therapy employs pharmacological agents that circulate throughout the bloodstream to eliminate rapidly proliferating cells body-wide?
a.) Chemotherapy b.) Surgery c.) Cryotherapy d.) Radiation therapy
- _____ 58. Which modern therapeutic approach boosts, trains, or genetically engineers a patient's immune system to recognize and eradicate malignant cells?
a.) Chemotherapy b.) Immunotherapy c.) Surgery d.) Radiation therapy
- _____ 59. What cancer treatment strategy starves growing tumors by administering drugs that block the signaling pathways responsible for recruiting new capillaries?
a.) Radiation therapy b.) Inhibition of blood vessel formation (anti-angiogenesis) c.) Chemotherapy d.) Gene therapy
- _____ 60. What emerging therapeutic paradigm seeks to modify, replace, or repair defective genetic sequences inside malignant cells to halt their proliferation or induce apoptosis?
a.) Gene therapy b.) Radiation therapy c.) Surgery d.) Chemotherapy

COMPLETE ANSWER KEY & EXPLANATIONS

Module 5: Genetics, Human Inheritance and Cancer — Reference Key with Concept Explanations

Item	Key	Correct Answer	Curriculum Explanation & Reference
1	C	Genetics	[Genetics Fundamentals] Genetics is the study of heredity in general and of genes in particular, forming a central pillar of biology.
2	B	Genes	[Genetics Fundamentals] Inheritance of traits is determined by factors passed to descendants, now designated as genes.
3	C	Alleles	[Alleles & Zygosity] Alleles are different forms of the same gene.
4	A	Homozygous	[Alleles & Zygosity] An individual with a pair of the same alleles is identified as homozygous.
5	C	Heterozygous	[Alleles & Zygosity] An individual with different alleles for a gene is referred to as heterozygous.
6	C	Dominant	[Alleles & Zygosity] In heterozygous individuals, the allele that is expressed is described as being dominant.
7	D	Recessive	[Alleles & Zygosity] The allele that is not expressed (masked) in a heterozygote is called recessive.
8	A	A nonfunctional protein or no protein at all	[Alleles & Zygosity] The dominant allele usually produces a functional protein, while the recessive allele produces either a nonfunctional protein or no protein at all.
9	B	Law of Segregation	[Mendelian Principles] The Law of Segregation states that allele pairs separate during gamete formation, with half receiving one allele and half the other.
10	C	Law of Independent Assortment	[Mendelian Principles] The Law of Independent Assortment dictates that genes on different chromosomes segregate independently into gametes.
11	C	Genotype	[Genotype vs Phenotype] The genotype refers to the specific alleles present and combined in an individual (e.g., AA, Aa, aa).
12	D	Phenotype	[Genotype vs Phenotype] The phenotype refers to the observable physical or physiological traits expressed in the individual.
13	B	Punnett square	[Genetic Crosses] A Punnett square is used to determine the probability that each gamete cross will occur.
14	C	100% Heterozygous (Aa)	[Genetic Crosses] Crossing AA x aa yields 100% heterozygous (Aa) offspring expressing the dominant phenotype.
15	B	25% AA, 50% Aa, 25% aa	[Genetic Crosses] An Aa x Aa cross produces 25% homozygous dominant (AA), 50% heterozygous (Aa), and 25% homozygous recessive (aa).

(aa).

16	B	75% dominant, 25% recessive	[Genetic Crosses] Because AA and Aa both show the dominant phenotype, 75% exhibit the dominant trait and 25% show the recessive trait.
17	D	Carrier	[Genetic Crosses] A heterozygous individual that shows the dominant phenotype while carrying a recessive allele is called a carrier.
18	C	Pedigree	[Pedigrees] Pedigrees are diagrams constructed to show genetic relationships among individuals in an extended family.
19	C	Codominance	[Non-Mendelian Inheritance] Codominance takes place when both alleles are fully and distinctly expressed in the phenotype (e.g., AB blood type).
20	A	Incomplete dominance	[Non-Mendelian Inheritance] Incomplete dominance occurs when expression in a heterozygote is intermediate between homozygous dominant and homozygous recessive.
21	B	Multiple alleles	[Non-Mendelian Inheritance] Multiple alleles refers to the condition where three or more alleles exist for a particular gene in a population, such as ABO blood types (IA, IB, i).
22	D	Polygenic inheritance	[Non-Mendelian Inheritance] Polygenic inheritance occurs when more than one gene controls the trait. Examples include human height, skin pigmentation, and eye color.
23	A	Sex-influenced traits	[Non-Mendelian Inheritance] The expression of sex-influenced traits depends on both the presence of the allele and circulating sex hormones (e.g. testosterone).
24	C	Free earlobes	[Human Genetic Traits] Definite free earlobes are dominant; attached earlobes are recessive.
25	A	Dominant	[Human Genetic Traits] A widow's peak is a dominant trait; a straight hairline is recessive.
26	B	Dominant	[Human Genetic Traits] Having six fingers (polydactyly) is genetically dominant; five fingers is recessive.
27	D	Left thumb over right thumb	[Human Genetic Traits] Placing the left thumb over the right thumb when crossing thumbs is dominant; right over left is recessive.
28	D	Dominant	[Human Genetic Traits] Presence of dimples is dominant; absence of dimples is recessive.
29	B	Dominant	[Human Genetic Traits] The ability to roll the lateral edges of the tongue upward is a dominant trait; inability is recessive.
30	A	Second toe longer than first	[Human Genetic Traits] Having the second toe longer than the big toe

		toe	(Morton's toe) is dominant; first toe longer is recessive.
31	C	Amniocentesis	[Detecting Genetic Disorders] In amniocentesis, a sample of amniotic fluid containing fetal cells is withdrawn from the uterus and analyzed for genetic problems.
32	D	Chorionic villi sampling (CVS)	[Detecting Genetic Disorders] Chorionic villi sampling (CVS) involves sampling tissue from the chorionic villi of the placenta for genetic diagnosis.
33	A	Uncontrolled cell division	[Cancer Biology] All cancers share one universal hallmark: uncontrolled and unregulated cell division.
34	A	Tumor	[Tumors & Neoplasms] A tumor (or neoplasm, meaning 'new growth') is an abnormal growth of cells that forms a tissue mass.
35	B	Benign tumor	[Tumors & Neoplasms] A benign tumor is surrounded by a connective tissue capsule and does not invade adjacent tissues or metastasize.
36	D	Malignant tumor	[Tumors & Neoplasms] Malignant tumors can invade neighboring tissues and colonize distant body locations; they are properly termed cancerous.
37	B	Metastasis	[Cancer Progression] Metastasis refers to the spread of cancer cells from one part of the body to another via the bloodstream or lymphatic system.
38	A	Dysplasia	[Stages of Cancer Development] Dysplasia describes abnormal changes in cell shape, nuclei, and internal tissue organization in precancerous tissue.
39	A	Carcinoma in situ	[Stages of Cancer Development] Carcinoma in situ is a localized tumor of roughly 1 million cells where the interior experiences nutrient starvation and waste poisoning.
40	D	Angiogenesis	[Cancer Progression] Cancer cells release chemicals that initiate the growth of new blood vessels, delivering nutrients, removing wastes, and providing a route for metastasis.
41	A	Apoptosis	[Control of Cancer] Apoptosis is programmed cell death triggered when genetic damage is too severe to be repaired.
42	A	Tumor suppressor genes	[Cancer Genetics] Tumor suppressor genes help detect damaged DNA, pause the cell cycle, and manage the repair of genetic defects.
43	D	Proto-oncogenes	[Cancer Genetics] Proto-oncogenes produce growth factors or signal transducers that stimulate normal, orderly cell division.
44	A	Oncogene	[Cancer Genetics] Mutation of a proto-oncogene results in an oncogene, which accelerates cell division without regulatory stimuli.
45	C	Natural killer cells and	[Immune Surveillance] Natural killer (NK) cells and cytotoxic T cells

		cytotoxic T cells	recognize and destroy abnormal or cancerous cells.
46	C	Carcinogen	[Cancer Causes] A carcinogen is an environmental agent that fosters cancer development by inducing genetic mutations.
47	A	Human Papilloma Virus (HPV)	[Cancer Causes] HPV is a DNA virus known to cause cervical and other genital/oropharyngeal cancers.
48	A	Hepatitis B and Hepatitis C viruses	[Cancer Causes] Chronic infections with Hepatitis B and C viruses significantly increase the risk of liver cancer.
49	D	Epstein-Barr virus (EBV)	[Cancer Causes] Epstein-Barr virus (EBV) is linked to infectious mononucleosis and specific malignancies such as Burkitt lymphoma.
50	D	Ultraviolet (UV) radiation	[Cancer Causes] Exposure to ultraviolet (UV) radiation causes pyrimidine dimers in DNA, elevating the risk of melanoma and other skin carcinomas.
51	D	Biopsy	[Diagnosing Cancer] A biopsy involves removing a small piece of tissue to examine whether cells display the characteristic architectural features of malignancy.
52	C	Tumor marker tests	[Diagnosing Cancer] Tumor marker tests detect circulating substances produced either directly by cancer cells or by host tissues reacting to a tumor.
53	B	Sputum	[Diagnosing Cancer] Sputum cytology and DNA testing are used to detect early signs of lung cancer.
54	D	Bladder cancer	[Diagnosing Cancer] Urine cytology and marker tests are utilized to detect signs of bladder cancer.
55	C	Colon cancer	[Diagnosing Cancer] Fecal immunochemical tests and stool DNA analyses are used to detect early signs of colorectal cancer.
56	D	Radiation therapy	[Treating Cancer] Radiation therapy causes extensive DNA damage within tumor cells, triggering programmed cell death.
57	A	Chemotherapy	[Treating Cancer] Chemotherapy drugs reach all parts of the body via the vascular system and selectively destroy rapidly dividing cells.
58	B	Immunotherapy	[Treating Cancer] Immunotherapy enhances the host immune response against cancer cells using cancer vaccines, checkpoint inhibitors, or cytokine factors.
59	B	Inhibition of blood vessel formation (anti-angiogenesis)	[Treating Cancer] Inhibition of blood vessel formation (anti-angiogenic therapy) prevents vascularization, depriving tumors of oxygen and nutrients.
60	A	Gene therapy	[Treating Cancer] Gene therapy alters the genetic material of cancer cells with the goal of repairing defective regulatory pathways or triggering

death.
